

3.1.1 Sequences

General Idea & Terminology:

We can think of a sequence as *an ordered list* of numbers. For example:

$$10, \quad 20, \quad 30, \quad 40, \quad 50, \quad \dots$$

NOTATION!

When talking about sequences (big lists of numbers) we usually write the following:

In general we will use a_n to represent the *general term* of a sequence by using a formula (like with our function above).

Example 1:

(a) Write the first 5 terms of the sequence given by $a_n = \frac{n}{n+1}$.

(b) Find the terms: a_3 , a_4 , and a_5 of the sequence given by $a_n = \sqrt{n-3}$ where $n \geq 3$

(c) Find the first five terms of the following sequence which is given *recursively* as:
 $f_1 = 1$ $f_2 = 1$ $f_n = f_{n-1} + f_{n-2}$ where $n \geq 3$. This sequence is known as the *Fibonacci sequence*.

Example 2 (you try): Find an explicit formula for the sequence whose first few terms are:

(a) $1, 8, 27, 64, 125, \dots$

(b) $1, 3, 7, 15, 31, 63, 127, \dots$

(c) $1, \frac{-1}{2}, \frac{1}{3}, \frac{-1}{4}, \frac{1}{5}, \frac{-1}{6}, \dots$

Question to the Class: If $a_n = \frac{n}{n+1}$, what do you think $\lim_{n \rightarrow \infty} a_n$ should equal?

Definition:

Note:

Example 3: Use the above definition to prove that if $a_n = \frac{n}{n+1}$, then $\lim_{n \rightarrow \infty} a_n = 1$.

Definition:

Note: $\lim_{n \rightarrow \infty} a_n = -\infty$ can be defined similarly.

Theorem:

Proof: The proof follows from the definition of limit of a function.

Example 4: Use the above theorem to show that if $a_n = \frac{n}{n+1}$, then $\lim_{n \rightarrow \infty} a_n = 1$.

Example 5 (you try): Find with full justification: $\lim_{n \rightarrow \infty} \frac{\ln(n)}{n}$.

The next theorem tells us that analogues of the limit laws we learned in Calculus I hold for sequences. They can be proven similarly to the limit laws presented in Calculus I.

Theorem: If $\{a_n\}$ and $\{b_n\}$ are convergent sequences and c is a constant, then

1. $\lim_{n \rightarrow \infty} (a_n \pm b_n) = \lim_{n \rightarrow \infty} a_n \pm \lim_{n \rightarrow \infty} b_n$
2. $\lim_{n \rightarrow \infty} c a_n = c \lim_{n \rightarrow \infty} a_n$
3. $\lim_{n \rightarrow \infty} (a_n \cdot b_n) = \lim_{n \rightarrow \infty} a_n \cdot \lim_{n \rightarrow \infty} b_n$
4. $\lim_{n \rightarrow \infty} \left(\frac{a_n}{b_n} \right) = \frac{\lim_{n \rightarrow \infty} a_n}{\lim_{n \rightarrow \infty} b_n}$ if $b_n \neq 0$
5. $\lim_{n \rightarrow \infty} a_n^p = \left[\lim_{n \rightarrow \infty} a_n \right]^p$ if $p > 0$ and $a_n > 0$.

Theorem:

Note: This is the analogue of the squeeze theorem and its proof is similar to the proof of the squeeze theorem.

Theorem:

Note:

Example 6:

Evaluate $\lim_{n \rightarrow \infty} \left(\frac{(-1)^n}{n} \right)$ if it exists.

Example 7:

Determine whether each sequence converges or diverges (we implicitly assume that this means as n goes to infinity). If it converges find the exact limit. If it diverges provide an explanation as to why it diverges.

(a) $a_n = (-1)^n$

(b)
$$b_n = \sqrt{\frac{n+1}{9n+1}}$$

(c)
$$c_n = \frac{(-1)^{n+1} n}{n + \sqrt{n}}$$

Theorem:

Proof: This fact follows from the epsilon delta definition of continuity (details will be omitted).

Example 8: Determine whether $a_n = \cos\left(\frac{2}{n}\right)$ is convergent or divergent.

Theorem:

Example 9: Determine (with justification) whether $a_n = \frac{9^{n-1}}{10^{2n+1}}$ converges or diverges.